

# BAULKHAM HILLS HIGH SCHOOL



## MATHEMATICS EXTENSION 2 ASSESSMENT

**December 2012**

*Time allowed: 50 minutes  
plus 5 minutes reading time*

**BOS NUMBER :** \_\_\_\_\_

**TEACHER'S NAME:** \_\_\_\_\_

| QUESTION          | MARK       |
|-------------------|------------|
| 1                 | /10        |
| 2                 | /10        |
| 3                 | /7         |
| 4                 | /11        |
| <b>TOTAL</b>      | <b>/38</b> |
| <b>PERCENTAGE</b> |            |



# Extension 2 Mathematics

December 2012

Time: 50 minutes + 5 minutes reading time

## DIRECTIONS

- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Use black or blue pen only (*not pencils*) to write your solutions.
- No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.
- Approved Maths aids and calculators may be used
- Marks are indicated next to each question

## QUESTION 1 Start on the appropriate page in your booklet (10 marks)

a If  $u = 1 - i$  and  $v = -2 + 4i$  evaluate the following

- i.  $u^2 + v$  1
- ii.  $\text{Im}(u + \bar{v})$  2
- iii.  $\left| \frac{u+v+1}{u+v+i} \right|$  2

b Let  $z$  the complex number  $z = \sqrt{3} + i$   
Express in mod-arg form (with principal argument)

- i.  $z$  2
- ii.  $iz$  1
- iii.  $z^9$  2

## QUESTION 2 Start on the appropriate page in your booklet (10 marks)

- a Find both square roots of  $-3 + 4i$  2
- b Sketch the locus of  $z$  if  $\arg(z - 1) = \arg(z - i)$  2
- c
  - i. Sketch  $|z - 2i| \leq 1$  2
  - ii. Find the largest value of  $\arg(z)$  when  $|z - 2i| = 1$  1
- d Sketch the locus of  $z$  if  $\text{Re}\left(\frac{-i}{z}\right) = 2 \text{Im}\left(-\frac{i}{z}\right)$  3

CONTINUED ON NEXT PAGE

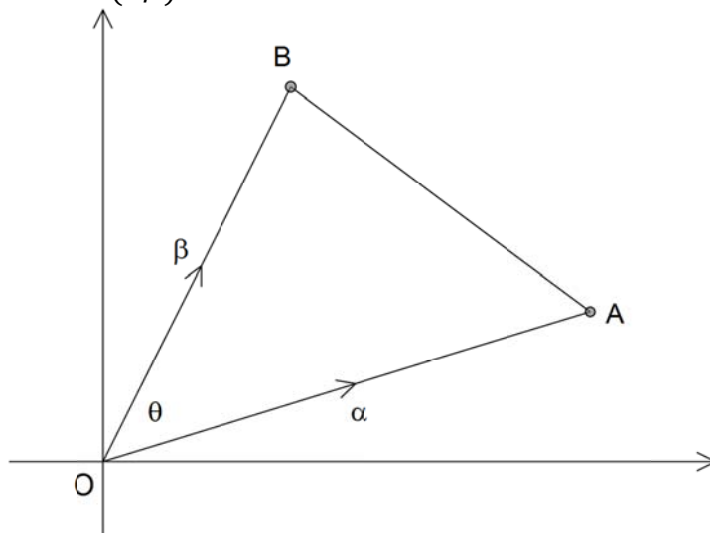
**QUESTION 3 Start on the appropriate page in your booklet (7 marks)**

- a**
- i. Given  $|z| = 1$ , show that  $z^n + z^{-n} = 2 \cos n\theta$ . 2
  - ii. Solve  $z^6 + 1 = 0$  and 3  
hence factorise  $z^6 + 1$  into real quadratic factors
  - iii. Hence deduce that  $\cos 3\theta = \cos \theta (2 \cos \theta + \sqrt{3})(2 \cos \theta - \sqrt{3})$  2

**QUESTION 4 Start on the appropriate page in your booklet (11 marks)**

- a** Given that  $z + \frac{1}{z} = k$  where  $k$  is real and  $z = x + iy$
- i. Show that either  $y = 0$  or  $x^2 + y^2 = 1$  2
  - ii. Show that if  $y = 0$  then  $|k| \geq 2$  2
  - iii. Show that if  $x^2 + y^2 = 1$  then  $|k| \leq 2$  2
- b**
- i. Use the results  $z + \bar{z} = 2\operatorname{Re}(z)$  and  $|z|^2 = z\bar{z}$  to show that 3  

$$|\alpha|^2 + |\beta|^2 - |\alpha - \beta|^2 = 2 \operatorname{Re}(\alpha\bar{\beta})$$
  - ii. The diagram shows the angle  $\theta$  between the complex numbers  $\alpha$  and  $\beta$  2  
 Prove that  $|\alpha\beta| \cos \theta = \operatorname{Re}(\alpha\bar{\beta})$



# Extension 2 December Task 2012

1 a) i)  $(1-i)^2 - 2+4i$   
 $= 1-2i+i^2-2+4i$   
 $= -2+2i$  ✓

ii)  $\text{Im}(1-i + (-2-4i))$  ① for conjugate

$= \text{Im}(-1-5i)$   
 $= -5$  ✓

iii)  $\frac{|1-i-2+4i+1|}{|1-i-2+4i+i|}$

$= \frac{|3i|}{|-1+4i|}$

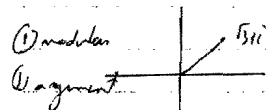
$= \frac{|3i|}{|-1+4i|}$

$= \frac{3}{\sqrt{1+4^2}}$

$= \frac{3}{\sqrt{17}}$  or  $\frac{3\sqrt{17}}{17}$  ✓

or equivalent:  $\frac{3i(-1-4i)}{(-1+4i)(-1-4i)}$  ✓

1 b) i)  $z = |z|(\cos 30^\circ + i \sin 30^\circ)$   
 $= \sqrt{1+1}(\cos 30^\circ + i \sin 30^\circ)$   
 $z = 2(\cos 30^\circ + i \sin 30^\circ)$  ✓



ii)  $iz = 2(\cos 120^\circ + i \sin 120^\circ)$  ✓

iii)  $z^9 = (2(\cos 30^\circ + i \sin 30^\circ))^9$

$z = 512(\cos 270^\circ + i \sin 270^\circ)$  ✓

$z = 512(\cos(-90^\circ) + i \sin(-90^\circ))$  ✓

2 a)  $x+iy = \sqrt{-3+4i}$

$x^2-y^2+2xyi = -3+4i$   
 by inspection  $\sqrt{-3+4i} = \pm(1+2i)$  ✓

(or  $x^2-y^2 = -3$   $2xy = 4$   
 $y = \frac{2}{x}$  ✓

$x^2 - \frac{4}{x^2} = -3$

$x^4 - 4 = -3x^2$

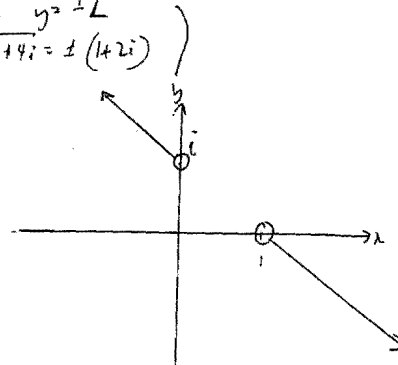
$x^4 + 3x^2 - 4 = 0$

$(x^2+4)(x^2-1) = 0$

no real soln  $x^2-1=0$   
 $x = \pm 1$

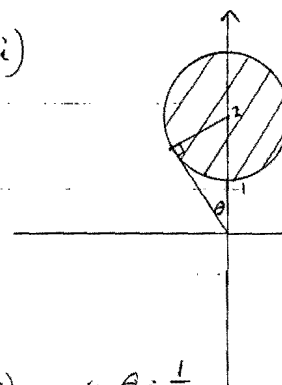
$\therefore y = \pm 2$   
 $\therefore \sqrt{-3+4i} = \pm(1+2i)$  ✓

b)



① endpoints ✓  
 ② line excluding interval from i to 1. ✓

c) i)



② circle, centre  $2i$ , with  $i$  or  $3i$  ✓  
 labelled, & shaded inside

① progress toward ✓  
 (eg centre  $2i$ , shaded inside but no clear radius)

ii)  $\sin \theta = \frac{1}{2}$   
 $\theta = 30^\circ$

$\therefore \text{max argument} = 120^\circ$  or  $(\frac{2\pi}{3})$  ✓

$$2d) \operatorname{Re}\left(\frac{-i}{z}\right) = 2 \operatorname{Im}\left(\frac{-i}{z}\right) \quad \text{NB } z \neq 0$$

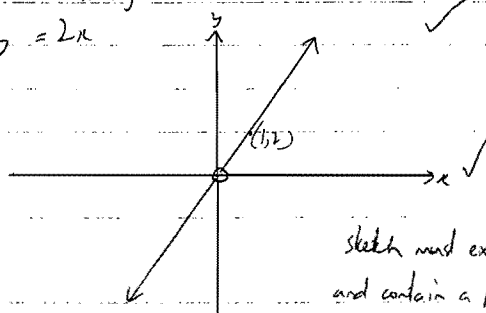
$$\text{let } z = x + iy$$

$$\operatorname{Re}\left(\frac{-i(x-iy)}{z\bar{z}}\right) = 2 \operatorname{Im}\left(\frac{-i(x-iy)}{z\bar{z}}\right)$$

$$\operatorname{Re}\left(\frac{-y-ix}{x^2+y^2}\right) = 2 \operatorname{Im}\left(\frac{-y-ix}{x^2+y^2}\right)$$

$$\frac{-y}{x^2+y^2} = \frac{2(-x)}{x^2+y^2}$$

$$y = 2x$$



sketch must exclude  $z=0$   
and contain a point/scale.

$$\begin{aligned} a) i) \text{ let } z &= 1(\cos \theta + i \sin \theta) \quad |z|=1 \\ z^n + z^{-n} &= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n} \\ &= \cos(n\theta) + i \sin(n\theta) + \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos(n\theta) + i \sin(n\theta) + \cos(n\theta) - i \sin(n\theta) \\ z^n + z^{-n} &= 2 \cos n\theta \quad \text{as req'd.} \end{aligned}$$

$$(ii) \quad z^6 = -1 \\ z_k = \operatorname{cis}\left(\frac{\pi + 2k\pi}{6}\right) \quad k=0, 1, 2, 3, 4, 5 \quad (\text{or equivalent})$$

$$z_0 = \operatorname{cis} \frac{\pi}{6}$$

$$z_1 = \operatorname{cis} \frac{3\pi}{6}$$

$$z_2 = \operatorname{cis} \frac{5\pi}{6}$$

$$z_3 = \operatorname{cis} \frac{7\pi}{6} = \bar{z}_2$$

$$z_4 = \operatorname{cis} \frac{9\pi}{6} = \bar{z}_1$$

$$z_5 = \operatorname{cis} \frac{11\pi}{6} = \bar{z}_0$$

$$k^2 - 4x + 1 \geq 0$$

$$k^2 - 4 \geq 0$$

$$k^2 \geq 4$$

$$|k| \geq 2$$

$$(iii) \quad \left|z + \frac{1}{z}\right| = |k|$$

$$\left|z + \frac{1}{z}\right| \geq |k|$$

$$\text{as } \left|z + \frac{1}{z}\right| \geq \left|z + \frac{1}{z}\right|$$

$$\left|z + \frac{1}{z}\right| \geq |k|$$

$$1 + \frac{1}{|z|} \geq |k|$$

$$\text{as } |z|=1 \text{ if } x^2+y^2=1$$

$$|k| \leq 2$$

(method 2: from above

$$(x+iy)(x^2+y^2) + x-iy = k(x^2+y^2)$$

$$x^2+iy^2 + x-iy = k$$

$$2x = k$$

$$(\text{as } x^2+y^2=1)$$

$$\text{Since } x^2+y^2=1 \quad -1 \leq x \leq 1$$

$$\therefore -2 \leq 2x \leq 2$$

$$-2 \leq k \leq 2$$

$$\text{ie } |k| \leq 2$$

$$b) i) \quad |\alpha|^2 + |\beta|^2 - |\alpha - \beta|^2$$

$$= \alpha\bar{\alpha} + \beta\bar{\beta} - (\alpha - \beta)(\bar{\alpha} - \bar{\beta})$$

$$= \alpha\bar{\alpha} + \beta\bar{\beta} - (\alpha\bar{\alpha} - \alpha\bar{\beta} - \beta\bar{\alpha} + \beta\bar{\beta})$$

$$= \alpha\bar{\alpha} + \beta\bar{\beta} - (\alpha\bar{\alpha} + \beta\bar{\beta} - \alpha\bar{\beta} - \bar{\alpha}\beta)$$

$$= \alpha\bar{\beta} + \bar{\alpha}\beta$$

$$= \alpha\bar{\beta} + \bar{\alpha}\beta$$

$$= 2 \operatorname{Re}(\alpha\bar{\beta})$$

$$z^4 + 1 = (z - z_1)(z - \bar{z}_1)(z - z_2)(z - \bar{z}_2)$$

$$= (z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1)(z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2)$$

$$= (z^2 - 2 \cos \frac{\pi}{2} z + 1)(z^2 - 2 \cos \frac{3\pi}{2} z + 1)$$

$$\text{ii)} \quad \frac{z^4 + 1}{z^3} = \frac{(z^2 + 1)(z^2 - \sqrt{3}z + 1)}{z} \quad \text{OR} \quad \left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}$$

$$z^3 + \frac{1}{z^3} = \left(z + \frac{1}{z}\right)\left(z + \frac{1}{z} - \sqrt{3}\right)\left(z + \frac{1}{z} + \sqrt{3}\right)$$

$$2 \cos 3\theta = 2 \cos \theta (2 \cos \theta - \sqrt{3})(2 \cos \theta + \sqrt{3})$$

$$\cos 3\theta = \cos \theta (2 \cos \theta - \sqrt{3})(2 \cos \theta + \sqrt{3})$$

$$\begin{aligned} (2 \cos \theta)^3 &= z^3 + \frac{1}{z^3} + 3\left(z + \frac{1}{z}\right) \\ 8 \cos^3 \theta &= 2 \cos 3\theta + 6 \cos \theta \\ \cos 3\theta &= 4 \cos^3 \theta - 3 \cos \theta \\ \cos 3\theta &= \cos \theta (4 \cos^2 \theta - 3) \\ \cos 3\theta &= \cos \theta (2 \cos \theta + \sqrt{3})(2 \cos \theta - \sqrt{3}) \end{aligned}$$

$$\text{Q4 a) i)} \quad z + \frac{1}{z} = k$$

$$x + iy + \frac{1}{x + iy} = k$$

$$x + iy + \frac{x - iy}{x^2 + y^2} = k$$

$$(x + iy)\left(\frac{x - iy}{x^2 + y^2}\right) + x - iy = k(x^2 + y^2) + 0i$$

$$\text{Equating imaginary parts } y(x^2 + y^2) - y = 0$$

$$y(x^2 + y^2 - 1) = 0$$

$$\therefore y = 0 \quad \text{or} \quad x^2 + y^2 - 1 = 0$$

$$\text{ie } x^2 + y^2 = 1$$

$$\text{ii)} \quad \text{If } y = 0 \quad z + \frac{1}{z} = k \text{ becomes}$$

$$x + \frac{1}{x} = k$$

$$\text{since } x \text{ and } k \text{ are real } \Delta \geq 0$$

$$x^2 + 1 = kx$$

$$x^2 - kx + 1 = 0$$

$$\text{b ii)} \quad \text{In } \triangle OBA$$

$$\cos \theta = \frac{|a|^2 + |b|^2 - |a - b|^2}{2|a||b|}$$

$$|\alpha||\beta| \cos \theta = \frac{2 \operatorname{Re}(\alpha \bar{\beta})}{2}$$

$$|\alpha \beta| \cos \theta = \operatorname{Re}(\alpha \bar{\beta})$$